

MODULE 1: WHAT IS WELDING & GAS CUTTING?

मॉड्यूल 1: वेल्डिंग एवं गैस कटिंग क्या है?

English

Welding is the process of joining metals using heat or electricity. Gas cutting is the process of cutting metal using oxygen and fuel gas.

हिन्दी

वेल्डिंग धातुओं को गर्मी या विद्युत की सहायता से जोड़ने की प्रक्रिया है। गैस कटिंग ऑक्सीजन और ईंधन गैस की सहायता से धातु को काटने की प्रक्रिया है।

MODULE 2: MAJOR HAZARDS

मॉड्यूल 2: प्रमुख खतरे

Hazards

- ✗ Fire
- ✗ Explosion
- ✗ Electric Shock
- ✗ Eye Injury
- ✗ Body Burn
- ✗ Gas Leakage
- ✗ Falling Spatters
- ✗ Toxic Fumes

Golden Rule

Every welding job is a hot work job.

हर वेल्डिंग कार्य एक हॉट वर्क है।

MODULE 3: WORK PERMIT REQUIREMENT

मॉड्यूल 3: वर्क परमिट

Before starting welding:

- ✓ Work Permit
- ✓ Hot Work Clearance
- ✓ Gas Clearance (if required)
- ✓ Confined Space Permit (if required)
- ✓ Height Work Permit (if required)

Rule

No Permit = No Welding

परमिट नहीं = वेल्डिंग नहीं

MODULE 4: FIRE PREVENTION

मॉड्यूल 4: अग्नि सुरक्षा

Before starting work:

- ✓ Remove combustible materials
- ✓ Check surroundings
- ✓ Arrange fire extinguishers
- ✓ Keep sand buckets ready
- ✓ Keep water buckets ready

Minimum Fire Protection

- ✓ 1 DCP Fire Extinguisher
- ✓ 2 Sand Buckets
- ✓ 2 Water Buckets

MODULE 5: FLASHBACK ARRESTOR

मॉड्यूल 5: फ्लैशबैक अरेस्टर

What is a Flashback?

Flame moving back into hose, regulator, or cylinder.

Why Flashback Arrestor?

- ✓ Stops flashback
- ✓ Prevents explosion
- ✓ Protects cylinders and hoses

Rule

Flashback arrestors must be installed on gas cutting systems.

MODULE 6: GAS CYLINDER SAFETY

मॉड्यूल 6: गैस सिलेंडर सुरक्षा

Do's

- ✓ Keep cylinders upright
- ✓ Secure cylinders with chain
- ✓ Use trolley for movement
- ✓ Keep safety caps installed
- ✓ Close valve after use

Don'ts

- ✗ Drop cylinders
- ✗ Drag cylinders
- ✗ Roll cylinders carelessly
- ✗ Use damaged cylinders

MODULE 7: OXYGEN & FUEL GAS CYLINDERS

मॉड्यूल 7: ऑक्सीजन एवं ईंधन गैस सिलेंडर

Important Rule

Maintain minimum 3 meters distance between oxygen and fuel gas cylinders.

Never

- ✗ Store cylinders together
- ✗ Keep cylinders near flame

✗ Take cylinders inside confined space

MODULE 8: HOSE SAFETY

मॉड्यूल 8: होज पाइप सुरक्षा

Colour Coding

● Red = Fuel Gas

● Black = Oxygen

Requirements

✓ Daily visual inspection

✓ No cracks

✓ No leakage

✓ Proper clamps

Never

✗ Use damaged hoses

✗ Use repaired hose without approval

MODULE 9: TORCH SAFETY

मॉड्यूल 9: टॉर्च सुरक्षा

Do's

✓ Check leaks before use

✓ Clean nozzle properly

✓ Use spark lighter

✓ Inspect every shift

Don'ts

✗ Light torch using matches

✗ Use damaged torch

MODULE 10: REGULATOR SAFETY

मॉड्यूल 10: रेगुलेटर सुरक्षा

Requirements

- ✓ Standard regulator
- ✓ Proper pressure gauge
- ✓ No leakage
- ✓ Proper identification

Never

- ✗ Use damaged regulator
- ✗ Use unapproved regulator

MODULE 11: ARC WELDING SAFETY

मॉड्यूल 11: आर्क वेल्डिंग सुरक्षा

Requirements

- ✓ Insulated electrode holder
- ✓ Proper welding cables
- ✓ Earthing
- ✓ Circuit protection
- ✓ Qualified welder

Hazards

- ✗ Electric shock
- ✗ Arc flash
- ✗ Burns

MODULE 12: WELDING MACHINE SAFETY

मॉड्यूल 12: वेल्डिंग मशीन सुरक्षा

Checks Before Use

- ☒ Cable condition
- ☒ Earthing
- ☒ Insulation condition
- ☒ Circuit breaker
- ☒ Proper connections

Never

- ☒ Use damaged cables
- ☒ Use exposed conductors
- ☒ Use faulty machine

MODULE 13: WELDING PPE

मॉड्यूल 13: वेल्डिंग PPE

Mandatory PPE:

- ☒ Welding Helmet
- ☒ Full Face Protection
- ☒ Leather Gloves
- ☒ Leather Apron
- ☒ Safety Shoes
- ☒ FR Clothing
- ☒ Safety Goggles

For Overhead Welding

- ☒ Shoulder Protection
- ☒ Leather Skull Cap
- ☒ Leg Protection

MODULE 14: WELDING IN CONFINED SPACE

मॉड्यूल 14: कन्फाईंड स्पेस में वेल्डिंग

Requirements:

- ☒ Confined Space Permit
- ☒ Ventilation
- ☒ Gas Monitoring
- ☒ Attendant Outside
- ☒ Rescue Plan
- ☒ Respiratory Protection if required

Never

- ☒ Weld without ventilation
- ☒ Weld without gas testing

MODULE 15: VENTILATION REQUIREMENTS

मॉड्यूल 15: वेंटिलेशन

Purpose

- ☒ Remove fumes
- ☒ Remove smoke
- ☒ Improve visibility
- ☒ Reduce exposure

Rule

Proper ventilation is mandatory during welding.

MODULE 16: OVERHEAD WELDING SAFETY

मॉड्यूल 16: ओवरहेड वेल्डिंग सुरक्षा

Before Work

- ☒ Follow work at height standard
- ☒ Proper lifting arrangement
- ☒ Barricade area below

- ✓ Prevent spatter fall
- ✓ Use spatter arrestor

Never

- ✗ Release structure only on tack welds
- ✗ Allow people below welding area

MODULE 17: WELDER QUALIFICATION

मॉड्यूल 17: वेल्डर योग्यता

Requirements

- ✓ Trained welder
- ✓ Certified welder
- ✓ Qualified for welding position

Overhead Welding

Only 4G or 6G qualified welder should perform overhead welding.

MODULE 18: HIGH PRESSURE WELDING

मॉड्यूल 18: हाई प्रेशर वेल्डिंग

For:

- ✓ Boiler Tubes
- ✓ HP Steam Lines
- ✓ Pressure Systems

Requirements

- ✓ IBR Approved Vendor
- ✓ Certified Welder
- ✓ Approved WPS
- ✓ Approved PQR
- ✓ Inspection & Testing

MODULE 19: SAFE OPERATION PRACTICES

मॉड्यूल 19: सुरक्षित संचालन

- ☒ Remove electrode when unattended
- ☒ Switch off machine when not in use
- ☒ Store cylinders safely
- ☒ Remove equipment after work
- ☒ Maintain housekeeping

MODULE 20: TEN GOLDEN RULES

मॉड्यूल 20: दस सुनहरे नियम

1. No Permit = No Welding
2. Use Flashback Arrestor
3. Use proper PPE
4. Keep fire extinguisher ready
5. Inspect hoses daily
6. Secure cylinders properly
7. Use proper ventilation
8. Maintain housekeeping
9. Use qualified welders
10. Stop work if unsafe